

Production functions



Economics has surprisingly little to say about the operation of firms: it considers them merely as entities that convert a set of inputs – capital, labour and skills – into output. The production function is a mathematical representation of this: it might indicate that if a textile factory had 10 looms and 10 workers it would produce 100 metres of cloth per day. It also tells you what happens when inputs change. For instance in this example inputs have to be deployed in fixed proportions: one worker to one loom. If the factory installs an extra loom, it will only increase output if it hires another worker to operate it.

Other kinds of production allow a choice between capital and labour: a farmer might choose to produce 10 tonnes of grain using tractors and a couple of drivers, or hiring many labourers to work by hand. The production function also describes what happens to output when inputs are scaled up. If inputs double

and output more than doubles – perhaps because scale allows specialization – the firm has ‘increasing returns to scale’.